中英对照。英文为原文，中文为译文。类名、参数名、函数名、资源路径与网址保持英文，便于在虚幻引擎细节面板中定位。插图、标题和条目顺序与原 PDF 一致。

Quickstart Tutorial

快速入门教程

The Vehicle Model

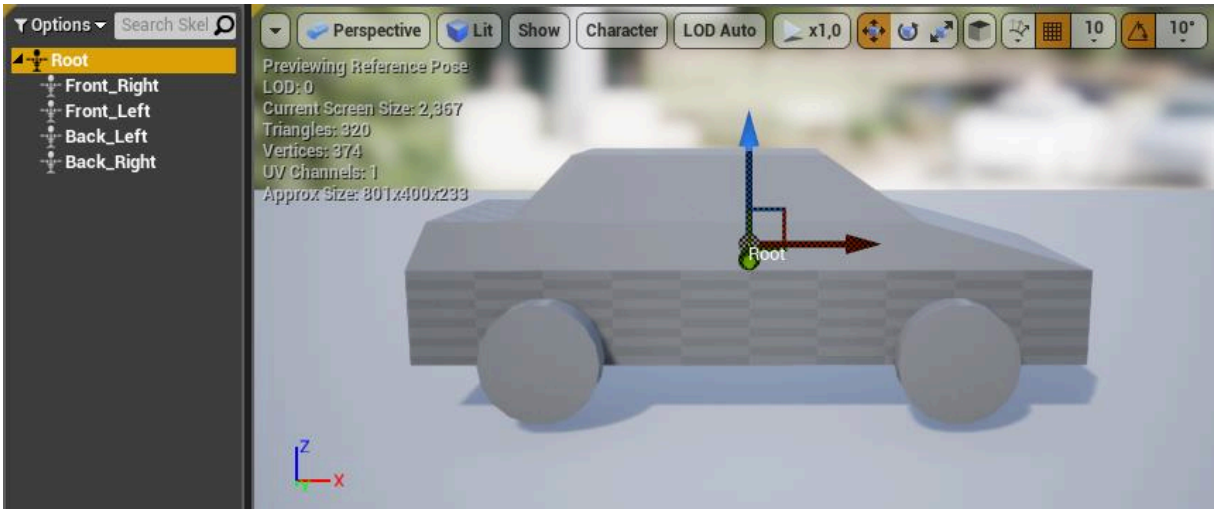
载具模型

FGear requires a rigged vehicle model. If your vehicle is ready for unreals own vehicle system then you can also use it with fgear. You can read the ue4 vehicle user guide, most concepts are similar : https://docs.unrealengine.com/en-US/Engine/Physics/Vehicles/VehicleUserGuide/index.html

FGear 需要一具已绑定骨骼的载具模型。如果该载具已经适配虚幻引擎自带的载具系统，也可以用于 FGear。可以阅读 UE4 载具用户指南，其中大部分概念是相近的：https://docs.unrealengine.com/en-US/Engine/Physics/Vehicles/VehicleUserGuide/index.html

Note : Your root bone must be located at zero origin(0,0,0) or you may encounter some problems.

注意：根骨骼必须位于零原点 (0,0,0)，否则可能会遇到一些问题。



The wheels should be in a normal ride height position.

车轮应处于正常行驶高度的位置。

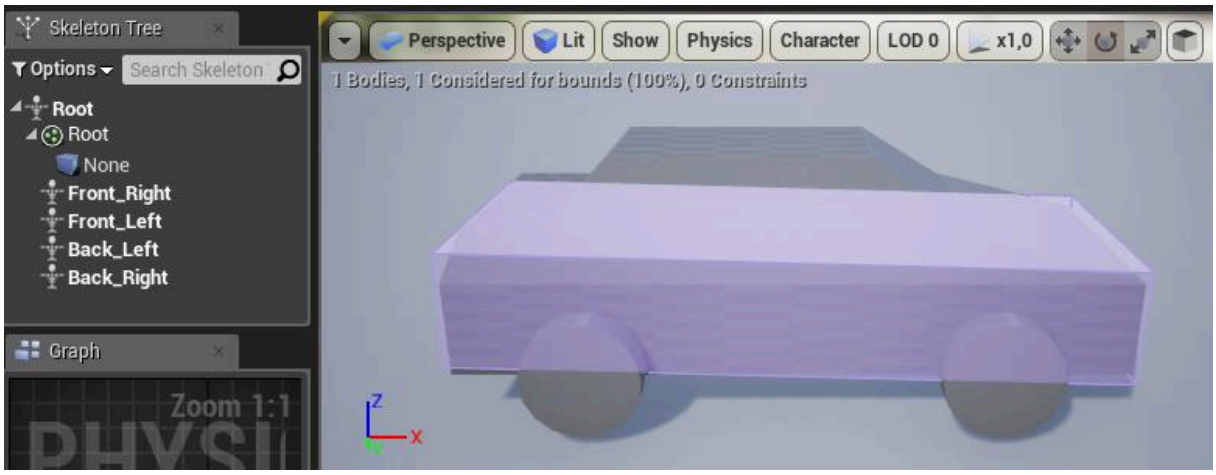
The Physics Asset

物理资产

You need to have a physics asset for your vehicle. This will represent your vehicles collision and inertial properties. You can also add a custom collision component for seperating collision and inertia. When custom collision is used this asset will only represent the inertial properties of the rigidbody. Check the extras section for custom collision component, for this tutorial a box collider will be sufficient.

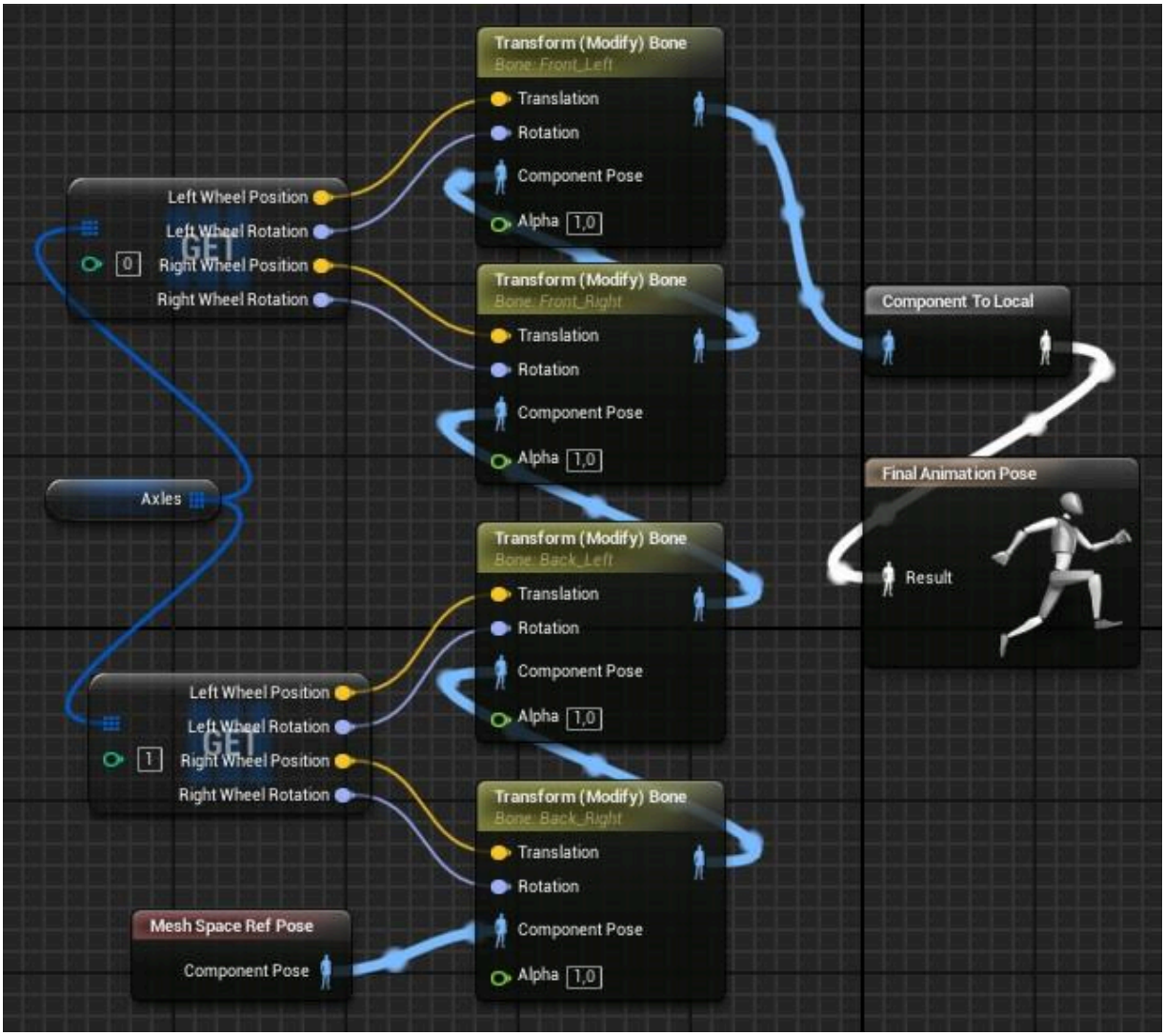
载具需要一份物理资产。它表示载具的碰撞与惯性属性。也可以添加自定义碰撞组件，把碰撞和惯性分开。使用自定义碰撞时，该资产只表示刚体的惯性属性。自定义碰撞组件见扩展部分；本教程使用一个盒体碰撞体即可。

Note : Starting with v1.2, physics substepping is supported and recommended.
注意：从 v1.2 起支持物理子步，并且推荐使用。



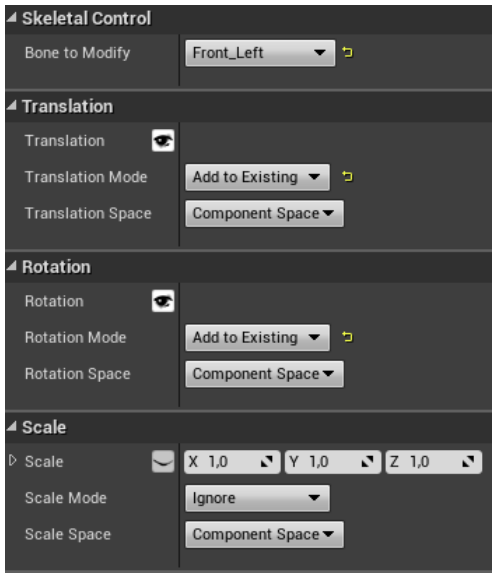
The Animation Blueprint
动画蓝图

Finally you need an animation blueprint for the vehicle. This will allow you to animate wheels transformation like spinning, turning and suspension travel. Use the “GetAxles” blueprint node to get each axle and the transform of left/right wheels and then set with the Transform Bone nodes. Here is an example image for a 4 wheel car:
最后还需要一份载具动画蓝图。它用来驱动车轮变换，例如滚动、转向和悬架行程。使用 “GetAxles” 蓝图节点取得每个车轴以及左/右车轮的变换，再用 Transform Bone 节点写回去。下面是四轮车的示例图：



And this is how each Transform Bone nodes are configured:

每个 Transform Bone 节点按如下方式设置：



Step by Step

逐步操作

1- Make sure FGear plugin is installed and activated. The plugin includes a sample car model, its physics asset and animation blueprint.

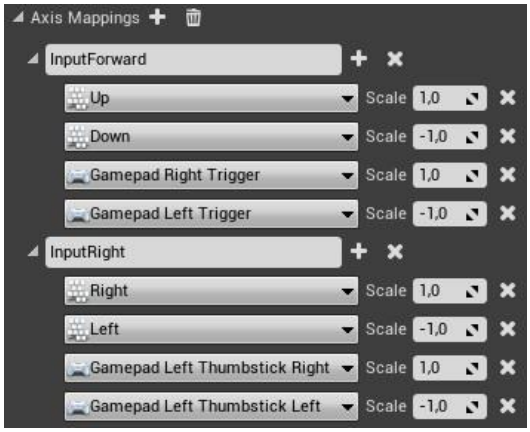
1- 确认 FGear 插件已安装并启用。插件包含一辆示例车模型，以及它的物理资产和动画蓝图。

2- Open a new default level and scale the floor to 100x, any extra content is optional.

2- 新建一个默认关卡，把地板缩放到 100 倍。其他额外内容可选。

3- Go to ProjectSettings->Input->Bindings and prepare two axis mappings namely "InputForward" and "InputRight" for throttle/brake and left/right steering:

3- 打开 Project Settings → Input → Bindings，准备两个轴映射，分别命名为 "InputForward" 和 "InputRight"，用于油门/制动以及左/右转向：



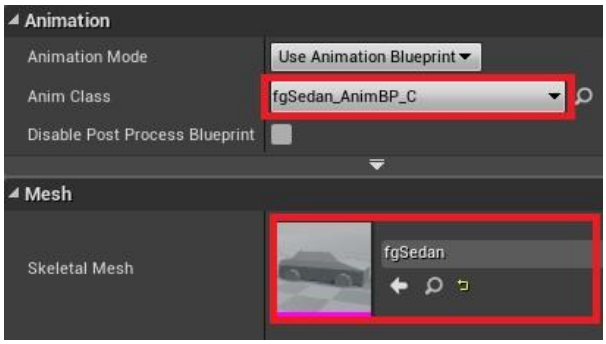
4- Create a new blueprint class, derive from FGearVehicle, name it as you like. Start editing the blueprint in full editor mode. All the necessary components are already created by default:

4- 新建一个蓝图类，继承 FGearVehicle，名称自定。用完整编辑器模式编辑该蓝图。所需组件默认都已创建：



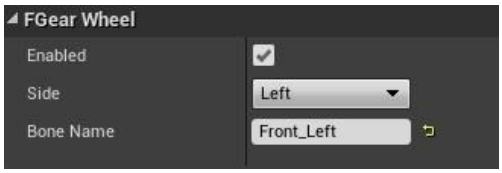
5- Select the mesh component(mMesh) and then set the skeletal mesh and the anim class. FGear contains a sample vehicle model fgSedan and animbp fgSedan_AnimBP:

5- 选中网格组件 (mMesh)，然后设置骨骼网格体和动画类。FGear 包含示例载具模型 fgSedan 和动画蓝图 fgSedan_AnimBP：



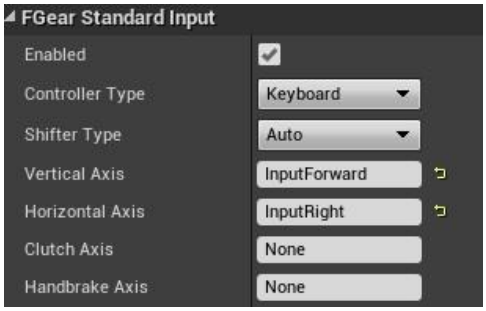
6- System requires bone locations so select each wheel(mWheelLeft0, mWheelRight0, mWheelLeft1, mWheelRight1) and set the BoneName parameters.

6- 系统需要骨骼位置，因此选中每个车轮（mWheelLeft0、mWheelRight0、mWheelLeft1、mWheelRight1），并设置 BoneName 参数。



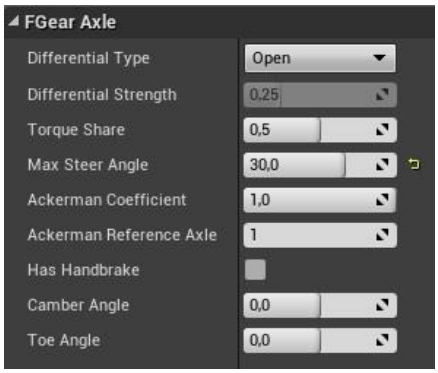
7- Select mStandardInput component and set VerticalAxis value to “InputForward” and HorizontalAxis value to “InputRight”.

7- 选中 mStandardInput 组件，把 VerticalAxis 设为 “InputForward”，把 HorizontalAxis 设为 “InputRight”。



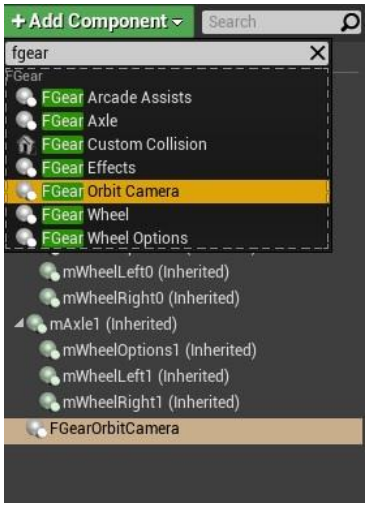
8- At this point vehicle is driveable but you can not steer yet so select the mAxle0 component and set the MaxSteerAngle to 30 degrees.

8- 此时载具已经可以行驶，但还不能转向。选中 mAxle0 组件，把 MaxSteerAngle 设为 30 度。



9- Finally add the FGearOrbitCamera component to the blueprint. You can also use a regular spring camera component.

9- 最后给蓝图添加 FGearOrbitCamera 组件。也可以使用普通的弹簧臂摄像机组件。



10- Compile and save the blueprint. Place the vehicle above the ground. Select the vehicle and set it as Player0. You should be able to drive around now. At this point all settings are default, you can start tweaking or take a look at the sample project.

10- 编译并保存蓝图。把载具放到地面上方。选中载具，将其设为 Player0。现在应当可以驾驶。此时全部设置为默认值，可以开始微调，或查看示例工程。

